Voice Activity Detection Using CLBP-Based Texture Encoding on Spectral Features

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Abstract. Developing lightweight yet robust voice activity detection (VAD) systems remains challenging under mismatched acoustic conditions. This study proposes a lightweight speech classification framework that combines handcrafted spectral representations, including Short-Time Fourier Transform (STFT), Mel-Frequency Cepstral Coefficients (MFCC), and Linear Frequency Cepstral Coefficients (LFCC), with Completed Local Binary Pattern (CLBP) encoding and conventional machine learning classifiers. The proposed framework is primarily evaluated for VAD on the MUSAN dataset, with additional cross-domain testing on Aurora-2 and transferability assessment on the ASVspoof 2019 Logical Access dataset. Experimental results show that XGBoost combined with STFT and CLBP achieves the best in-domain VAD performance on MUSAN, reaching 99.802% accuracy, 99.804% F1-score, an Equal Error Rate (EER) of 0.194% and Inference Time of 4.75ms. Cross-domain evaluation on Aurora-2 demonstrates that STFT-based representations provide more consistent robustness under unseen noisy acoustic conditions. Furthermore, experiments on ASVspoof 2019 Logical Access reveal that LFCC-based representations are more effective for spoof speech detection, achieving 96.12% accuracy and a 5.05% EER. These findings indicate that the proposed CLBP-enhanced framework is not only effective for lightweight VAD, but also transferable to related speech classification tasks.

Keywords: Voice Activity Detection, Completed Local Binary Pattern, Spectral Feature Representation, Texture-Based Feature Extraction, Machine Learning

1 Introduction

Voice Activity Detection (VAD) is an essential component in many speech processing systems, including automatic speech recognition, speaker verification, speech enhancement, and voice assistants. Its objective is to distinguish speech from non-speech regions such as silence and background noise. Although deep learning methods have recently achieved strong performance, developing lightweight and computationally efficient VAD systems remains a significant challenge, particularly for real-time and resource-constrained applications.

Traditional lightweight VAD systems commonly rely on handcrafted spectral representations combined with machine learning classifiers. Among the most widely used features, Short-Time Fourier Transform (STFT)[8], Mel-Frequency Cepstral Coefficients (MFCC)[4], and Linear Frequency Cepstral Coefficients (LFCC)[5] effectively characterize speech spectral information while maintaining low computational complexity. When combined with lightweight classifiers such as Support Vector Machine (SVM), Random Forest (RF), and XGBoost, these handcrafted features often provide competitive performance with reduced computational cost [1,2,3].

However, handcrafted spectral representations mainly focus on global spectral information and may insufficiently capture local structural patterns in speech signals. Since speech spectrograms exhibit structured two-dimensional patterns, texture analysis techniques originally developed for computer vision can be applied to extract discriminative local information. Among these methods, Local Binary Pattern (LBP) is a computationally efficient descriptor for encoding local neighborhood relationships [9]. To improve its representation capability, Completed Local Binary Pattern (CLBP) extends LBP by jointly encoding sign, magnitude, and center information, providing richer local texture characterization while maintaining low complexity [10]. Previous studies have shown the effectiveness of texture descriptors for spectrogram-based audio analysis [11].

Another challenge in practical VAD systems is domain shift, where acoustic conditions differ between training and testing environments due to variations in speakers, recording devices, and background noise. Consequently, models trained on one dataset may exhibit performance degradation when evaluated on unseen domains, making cross-domain generalization an important consideration for robust lightweight VAD. Despite growing interest in texture-based audio analysis, the effectiveness of CLBP across different handcrafted spectral representations for VAD remains insufficiently explored. In particular, systematic evaluation of CLBP-enhanced STFT, MFCC, and LFCC features under cross-domain settings is still limited.

Therefore, this study proposes a computationally efficient VAD framework integrating handcrafted spectral representations with CLBP-based local texture encoding. The main contributions of this study are as follows:

- A CLBP-enhanced lightweight VAD framework is proposed by combining handcrafted spectral representations with conventional machine learning classifiers.
- A comparative evaluation is conducted across STFT, MFCC, and LFCC representations using multiple machine learning models.
- Cross-domain robustness is assessed under mismatched acoustic conditions, and transferability is further examined on a related speech classification task.

The remainder of this paper is organized as follows. Section 2 presents the proposed methodology, Section 3 describes the experimental setup, Section 4 discusses the results, and Section 5 concludes the paper.

2 Methodology

2.1 Overview of the proposed framework

Figure 1 illustrates the overall architecture of the proposed lightweight speech classification framework, with voice activity detection as the primary task. Initially, the MUSAN corpus undergoes preprocessing and is divided via a K-fold cross-validation protocol to establish distinct training and testing partitions for in-domain analysis.

Within each training iteration, raw audio segments are transformed into different spectral representations, including STFT spectrograms as well as MFCC and LFCC feature matrices. Completed Local Binary Pattern (CLBP) encoding is then applied to these representations to capture local structural and textural patterns. The resulting histogram descriptors are subsequently used to train ML methods, including XGBoost, Random Forest, SVM, KNN, and Logistic Regression.

In the final phase, the trained models are evaluated on the held-out MUSAN test folds as well as independent external benchmarks. The Aurora-2 corpus is used to assess cross-domain robustness under mismatched noisy acoustic conditions. In addition, the ASVspoof 2019 LA dataset is employed to investigate whether the proposed CLBP-enhanced feature representations retain discriminative capability in a related but distinct speech classification task, namely bona fide versus spoofed speech detection. Performance is evaluated using standard metrics, including Accuracy, F1-score, Recall, and Equal Error Rate (EER), depending on the evaluation scenario.

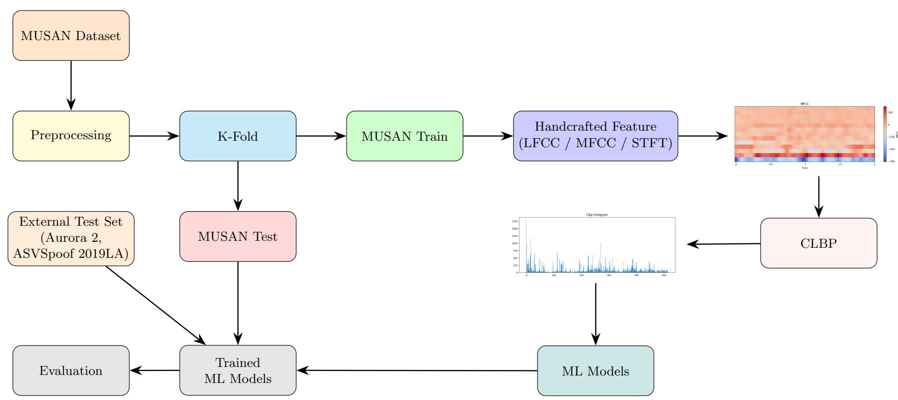

Fig. 1: Proposed lightweight VAD framework

2.2 Data Preprocessing

Before feature extraction, the raw audio signals undergo a two-stage preprocessing phase: silence removal and fixed-duration segmentation. To eliminate low-energy and non-informative regions, each audio signal is first partitioned into overlapping frames using a 25 ms window size with a 15 ms overlap. The short-time energy for each frame is computed as:

$$E = \sum_{n=0}^{N-1} x[n]^2$$

where $x[n]$ denotes the discrete audio samples within a frame of length N. Frames with energy values below a predefined threshold are discarded. To ensure consistent input length, the remaining audio is segmented into uniform 2-second clips for subsequent analysis. Each resulting clip is assigned to either the speech or non-speech category according to its source label.

2.3 Feature Extraction Methods

This study utilizes three handcrafted acoustic features: MFCC, LFCC, and STFT, which provide complementary cepstral and time-frequency information for speech analysis. CLBP encoding is then applied to the extracted feature matrices to enhance local structural and textural representations. Furthermore, to investigate the impact of feature dimensionality, MFCC and LFCC are evaluated across 20, 40, and 80 coefficients, while STFT is analyzed using a ‘FFT size‘ equal 257.

2.3.1 Mel-Frequency Cepstral Coefficients (MFCC) MFCC is a widely used acoustic feature representation that approximates human auditory perception by mapping the linear frequency spectrum onto the Mel scale. The extraction process involves applying STFT to the input signal, followed by Mel filter bank analysis, logarithmic energy computation, and a Discrete Cosine Transform (DCT). The resulting cepstral coefficients effectively capture the spectral envelope of speech signals [4].

2.3.2 Linear Frequency Cepstral Coefficients (LFCC) LFCC is extracted similarly to MFCC but uses linearly spaced filter banks instead of the Mel scale, preserving spectral information more uniformly. This enables finer representation of high-frequency components. Such characteristics make LFCC useful for tasks like speaker verification and spoof speech detection, where subtle spectral variations provide discriminative cues [5].

2.3.3 Short-Time Fourier Transform (STFT) STFT is a time-frequency analysis method commonly used for non-stationary signals such as speech. It divides the input waveform into short overlapping frames and applies the Fourier transform to each segment, generating a spectrogram that captures temporal variations in frequency content [8].

2.3.4 Completed Local Binary Pattern (CLBP) CLBP is an extension of the Local Binary Pattern (LBP) descriptor designed to provide a more complete representation of local texture information [9]. Unlike conventional LBP, which only encodes the sign of intensity differences between a central pixel and its neighboring pixels, CLBP additionally incorporates magnitude information and the gray-level intensity of the center pixel. Specifically, CLBP consists of three complementary descriptors: CLBP_S (sign), CLBP_M (magnitude), and CLBP_C (center intensity). By combining these descriptors and constructing histogram-based representations, CLBP provides richer local structural and textural information compared to traditional LBP [10].

$$\mathrm{CLBP\_S}_{P,R}(x_c) = \sum_{p=0}^{P-1} s(x_p - x_c)\, 2^p \tag{1}$$

$$\mathrm{CLBP\_M}_{P,R}(x_c) = \sum_{p=0}^{P-1} s(m_p - c)\, 2^p, \quad m_p = |x_p - x_c|, \quad c = \frac{1}{P}\sum_{p=0}^{P-1} m_p \tag{2}$$

$$\mathrm{CLBP\_C}(x_c) = s(x_c - \mu), \quad \mu = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{3}$$

$$\mathbf{Feature} = [\mathbf{Hist}_S, \mathbf{Hist}_M, \mathbf{Hist}_C] \tag{4}$$

In this study, each input audio waveform is first transformed into handcrafted spectral representations using MFCC, LFCC, or STFT. Each audio segment is represented as a two-dimensional feature matrix, where rows correspond to temporal frames and columns correspond to cepstral coefficients (MFCC/LFCC) or frequency bins (STFT), as illustrated in Fig. 2.

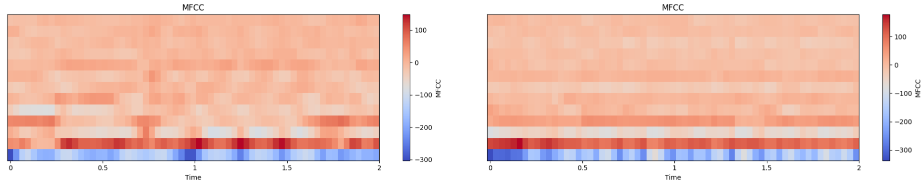

Fig. 2: Example spectral feature matrices (left: speech, right: non-speech)

CLBP encoding is then applied to the extracted feature matrices to capture local structural and textural patterns. In this study, the CLBP operator is configured with $P = 8$ sampling points and radius $R = 1$, following a commonly adopted configuration that provides a practical balance between discriminative capability and computational efficiency. The resulting binary patterns are summarized using histogram representations. Specifically, CLBP_S and CLBP_M each produce 256-bin histograms, while CLBP_C contributes a 2-bin histogram, yielding a final 514-dimensional feature vector for classification, as shown in Fig. 3.

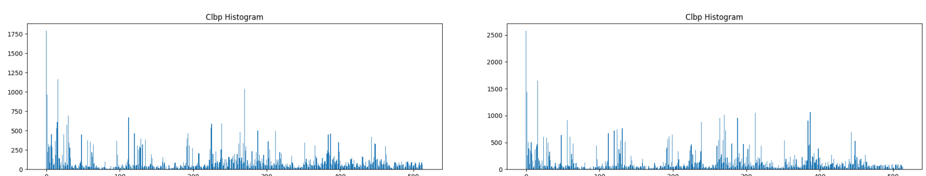

Fig. 3: CLBP histogram features (left: speech, right: non-speech)

Table 1: Feature extraction time for handcrafted features (sample's length: 2 second).

Feature	**20**	**40**	**80**	**257**
MFCC	2.2 ms	2.6 ms	3.5 ms	–
LFCC	3.9 ms	4.3 ms	6.0 ms	–
STFT	–	–	–	4.2 ms

2.4 Machine Learning Models

To assess the discriminative power of the extracted features, we utilize five traditional machine learning algorithms: K-Nearest Neighbors (KNN), Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). This selection covers a diverse range of learning paradigms, encompassing distance-centric, linear, margin-maximizing, and ensemble methodologies.

2.4.1 Extreme Gradient Boosting (XGBoost) As an advanced gradient boosting architecture, XGBoost constructs decision trees in a sequential manner, where each new iteration aims to minimize the residual errors of its predecessors. By integrating built-in regularization and rapid optimization, it delivers highly scalable and accurate predictive performance [16].

2.4.2 Random Forest (RF) Utilizing the bagging methodology, RF builds an ensemble of decision trees, each trained on random subsets of data and features. The aggregate prediction, determined by a majority vote, inherently mitigates the risk of overfitting and enhances overall model stability [15].

2.4.3 K-Nearest Neighbors (KNN) Operating as an instance-based, distance-driven method, KNN determines class membership based on the proximity of surrounding data points in the feature space. While conceptually straightforward, it yields competitive results provided the underlying feature representations are sufficiently distinct [12].

2.4.4 Logistic Regression (LR) Serving as a robust linear baseline, LR computes the probability of class assignment utilizing a logistic sigmoid function. Its widespread adoption in binary classification tasks stems primarily from its high interpretability and minimal computational footprint [13].

2.4.5 Support Vector Machine (SVM) is a supervised algorithm that seeks to identify an optimal hyperplane by maximizing the separation margin between distinct categories. It is highly proficient in managing high-dimensional data and leverages kernel functions to resolve non-linear boundaries effectively [14].

3 Experiments

3.1 Datasets

The proposed framework is primarily evaluated on the MUSAN corpus [6], a widely used benchmark for speech-related audio classification. MUSAN contains approximately 60 hours of speech recordings, 42 hours of music, and 7 hours of background noise. For binary classification, speech recordings are treated as the speech class, while music and noise samples are merged into the non-speech class. After preprocessing and segmentation, the resulting dataset contains 64,017 speech samples and 62,696 non-speech samples.

To evaluate cross-domain robustness under mismatched acoustic conditions, the Aurora-2 database [7] is used as an external test set. Aurora-2 consists of clean speech utterances corrupted with various real-world noise types across multiple signal-to-noise ratio (SNR) levels, making it suitable for assessing robustness under noisy environments.

To further investigate the transferability of the proposed feature representations, the ASVspoof 2019 Logical Access (LA) dataset [17] is additionally considered. This dataset contains 11.8 hours of bona fide speech and 97.8 hours of spoofed audio generated using voice conversion (VC) and text-to-speech (TTS) techniques. Unlike MUSAN and Aurora-2, this dataset represents a related but distinct speech classification task, enabling evaluation of whether the proposed representations remain effective beyond conventional VAD scenarios.

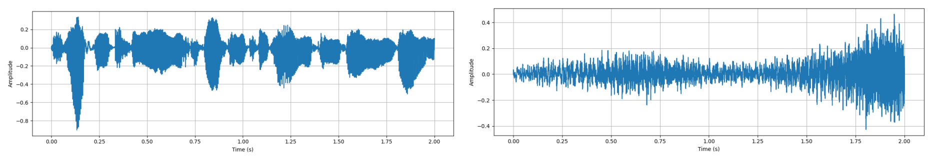

Fig. 4: Example waveform segments of speech and non-speech audio (on Musan corpus[6])

3.2 Performance Evaluation

To ensure a robust assessment of the proposed methodology, we implemented a 5-fold cross-validation strategy on the MUSAN corpus. The data was partitioned into five mutually exclusive subsets. During each validation cycle, four partitions served as the training data while the single hold-out partition was utilized for testing. The ultimate performance metrics were derived by calculating the mean across all five iterations.

The classification efficacy was quantified using three standard evaluation criteria: Accuracy, F1-score, and Equal Error Rate (EER). Accuracy reflects the global percentage of correct classifications. The F1-score computes the harmonic mean of Precision and Recall, offering a comprehensive view of the model's reliability in making positive predictions and its sensitivity in correctly identifying actual speech segments.

Furthermore, the Equal Error Rate (EER) was adopted to analyze the balance between false rejections and false acceptances. EER identifies the specific operating threshold where the False Positive Rate (FPR) strictly matches the False Negative Rate (FNR). A minimized EER signifies a superior capacity to differentiate between vocal and non-vocal audio classes. The mathematical formulation of EER is expressed as (5):

$$\mathrm{EER} = \mathrm{FPR}(\tau^*) \quad \text{where} \quad \tau^* = \arg\min_{\tau} |\mathrm{FPR}(\tau) - \mathrm{FNR}(\tau)| \tag{5}$$

4 Results and Discussion

This section presents the experimental results for evaluating the proposed framework on voice activity detection, including both in-domain and cross-domain settings, as well as its transferability to a related speech classification task.

4.1 Performance on Voice Activity Detection

4.1.1 In-domain Evaluation on MUSAN Table 2 summarizes the performance of different machine learning models on the MUSAN dataset using MFCC, LFCC, and STFT representations, alongside their corresponding inference times. Overall, most configurations achieved strong performance, with many settings exceeding 99% accuracy, indicating that the proposed CLBP-enhanced framework is highly effective for speech/non-speech classification under in-domain conditions.

Among the evaluated feature representations, STFT generally achieved the strongest and most consistent performance across different classifiers. In particular, XGBoost combined with STFT (257 components) achieved the best overall performance, reaching 99.8027% accuracy, 99.8048% F1-score, and the lowest EER of 0.1946%. This may be attributed to STFT's ability to preserve detailed time-frequency information directly from the original waveform, providing richer local spectral structures for CLBP encoding compared with cepstral representations. Notably, MFCC (40 coefficients) also showed highly competitive results when paired with XGBoost, yielding 99.8019% accuracy and a 0.1962% EER.

A clear performance trend can also be observed across feature types. STFT consistently outperformed LFCC and generally achieved stronger or comparable performance relative to MFCC. In contrast, LFCC-based configurations tended to produce lower accuracy and higher EER, particularly for SVM, KNN, and LR. This suggests that while LFCC may retain useful spectral details, its linear frequency emphasis may be less effective for distinguishing speech from non-speech content in the VAD setting, where broader spectral-temporal energy patterns appear to be more informative.

From the classifier perspective, tree-based ensemble methods, particularly XGBoost, consistently demonstrated strong performance while maintaining excellent computational efficiency. The newly introduced inference time evaluation reveals a significant advantage for XGBoost and LR. While LR achieved the absolute fastest inference times (e.g., 0.37 ms with MFCC-20), XGBoost remained remarkably fast at just 0.55 ms when paired with the optimal STFT features. In contrast, KNN was orders of magnitude slower (ranging from approximately 54 ms to 105 ms) due to its lazy-learning distance computations. This demonstrates that XGBoost not only effectively exploits discriminative non-linear interactions within the extracted feature space but also operates with extremely low latency, making it highly suitable for real-time VAD applications.

Another notable observation is that increasing feature dimensionality did not always lead to performance improvements. For example, MFCC-40 slightly outperformed MFCC-80 in several configurations, despite the larger feature dimensionality. This indicates that additional coefficients may introduce redundant or less informative components, which do not necessarily improve classifier performance and may even reduce robustness.

Finally, the relatively small standard deviations across repeated folds indicate stable performance and suggest that the proposed framework maintains consistent behavior under different train-test splits within the MUSAN corpus.

Figure 5 further illustrates the distribution of the STFT-CLBP feature representations after UMAP-based dimensionality reduction. The visualization shows that speech and non-speech samples form largely separable clusters, although partial overlap remains in certain regions of the feature space. Notably, speech samples form several relatively compact clusters, whereas non-speech samples exhibit a broader and more dispersed distribution, which may reflect the greater diversity of the non-speech category, including both music and background noise. This visualization suggests that the proposed STFT-CLBP representation cap-

Table 2: Performance of XGBoost on the MUSAN dataset (%, red: the best, cyan: second best)

Model	Feature	Coef	Accuracy	F1	EER	Inf. Time
XGBoost	MFCC	20	99.6851 ± 0.0379	99.6885 ± 0.0375	0.3206 ± 0.0332	0.43 ms
	MFCC	40	99.8019 ± 0.0386	99.8040 ± 0.0382	0.1962 ± 0.0424	0.60 ms
	MFCC	80	99.7325 ± 0.0506	99.7354 ± 0.0499	0.2552 ± 0.0413	0.603 ms
	LFCC	20	99.2629 ± 0.0419	99.2713 ± 0.0413	0.7385 ± 0.0444	0.87 ms
	LFCC	40	99.5075 ± 0.0623	99.5132 ± 0.0616	0.4961 ± 0.0747	0.69 ms
	LFCC	80	99.6859 ± 0.0291	99.6894 ± 0.0287	0.3126 ± 0.0362	0.69 ms
	STFT	257	99.8027 ± 0.0225	99.8048 ± 0.0223	0.1946 ± 0.0279	0.55 ms
RF	MFCC	20	99.12 ± 0.0676	99.13 ± 0.0676	0.82 ± 0.0379	1.47 ms
	MFCC	40	99.56 ± 0.0344	99.57 ± 0.0342	0.42 ± 0.0313	2.58 ms
	MFCC	80	99.50 ± 0.0504	99.51 ± 0.0499	0.48 ± 0.0643	1.48 ms
	LFCC	20	98.70 ± 0.0665	98.72 ± 0.0659	1.30 ± 0.0808	2.56 ms
	LFCC	40	98.96 ± 0.0652	98.97 ± 0.0647	1.04 ± 0.0518	2.58 ms
	LFCC	80	99.27 ± 0.0786	99.28 ± 0.0779	0.74 ± 0.0762	1.66 ms
	STFT	257	99.49 ± 0.0213	99.50 ± 0.0211	0.50 ± 0.0167	1.96 ms
SVM	MFCC	20	98.75 ± 0.0570	98.77 ± 0.0568	1.26 ± 0.0508	1.19 ms
	MFCC	40	99.28 ± 0.0506	99.29 ± 0.0500	0.72 ± 0.0594	1.68 ms
	MFCC	80	99.21 ± 0.0466	99.22 ± 0.0461	0.81 ± 0.0534	1.46 ms
	LFCC	20	98.13 ± 0.0370	98.15 ± 0.0365	1.88 ± 0.0458	1.50 ms
	LFCC	40	98.48 ± 0.1114	98.49 ± 0.1093	1.53 ± 0.1254	2.05 ms
	LFCC	80	99.20 ± 0.0495	99.21 ± 0.0485	0.79 ± 0.0493	2.00 ms
	STFT	257	99.53 ± 0.0316	99.53 ± 0.0313	0.48 ± 0.0307	1.32 ms
KNN	MFCC	20	97.87 ± 0.1418	97.92 ± 0.1365	1.77 ± 0.1314	87.0 ms
	MFCC	40	99.37 ± 0.0589	99.38 ± 0.0579	0.55 ± 0.0396	78.23 ms
	MFCC	80	99.51 ± 0.0825	99.51 ± 0.0811	0.43 ± 0.0657	81.3 ms
	LFCC	20	95.58 ± 0.1083	95.58 ± 0.1067	4.22 ± 0.1056	105.3 ms
	LFCC	40	95.90 ± 0.1219	95.82 ± 0.1274	3.14 ± 0.1436	88.0 ms
	LFCC	80	98.96 ± 0.0635	98.97 ± 0.0625	1.09 ± 0.0687	83.9 ms
	STFT	257	99.51 ± 0.0276	99.51 ± 0.0272	0.46 ± 0.0402	54.6 ms
LR	MFCC	20	98.75 ± 0.0550	98.77 ± 0.0549	1.26 ± 0.0467	0.37 ms
	MFCC	40	99.28 ± 0.0475	99.29 ± 0.0468	0.71 ± 0.0430	0.93 ms
	MFCC	80	99.20 ± 0.0544	99.21 ± 0.0540	0.80 ± 0.0511	0.85 ms
	LFCC	20	98.13 ± 0.0263	98.15 ± 0.0258	1.88 ± 0.0308	0.85 ms
	LFCC	40	98.46 ± 0.1436	98.47 ± 0.1412	1.53 ± 0.1458	0.77 ms
	LFCC	80	99.19 ± 0.0386	99.20 ± 0.0381	0.83 ± 0.0476	1.00 ms
	STFT	257	99.50 ± 0.0278	99.50 ± 0.0276	0.51 ± 0.0265	0.44 ms

tures discriminative structural and textural patterns that facilitate separation between the two classes, which is consistent with the strong classification performance observed in Table 2, particularly for ensemble-based classifiers such as XGBoost and RF.

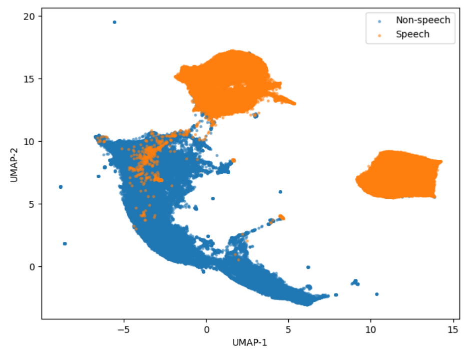

Fig. 5: STFT Feature Distribution for Speech and Non-Speech (using UMAP)

4.1.2 Cross-domain Evaluation on Aurora-2 Table 3 summarizes the cross-domain recall performance of models trained on MUSAN and evaluated on the Aurora-2 dataset. Recall is emphasized in this experiment because sensitivity to speech detection is particularly important under mismatched noisy conditions, where missed speech segments can significantly affect downstream applications.

A notable performance shift is observed compared with the in-domain MUSAN results, reflecting the impact of domain mismatch between training and testing conditions. Although KNN with MFCC-20 achieved the highest single recall (88.12%), this performance was not consistently maintained across other MFCC configurations or classifiers. In contrast, STFT demonstrated stronger overall robustness, achieving the highest recall for XGBoost (84.05%), SVM (66.38%), and LR (73.77%), while maintaining competitive performance for RF and KNN.

This suggests that STFT-based representations generalize more consistently under unseen noisy acoustic conditions, likely due to their ability to preserve richer time–frequency information relevant to speech activity detection. The relatively unstable performance of MFCC and LFCC across classifiers may indicate greater sensitivity to domain mismatch, particularly when feature representations become more task- or dataset-specific.

While certain low-dimensional MFCC configurations may generalize well in specific cases, STFT provides the most reliable cross-domain performance, supporting its suitability for robust voice activity detection.

Table 3: Performance of Machine Learning Models on the AURORA 2.0 dataset (%). Since the dataset contains only speech samples, Recall is used as the sole evaluation metric.

	Feature	**numCoef**	**XGBoost**	**RF**	**SVM**	**KNN**	**LR**
Recall %	MFCC	20	68.26	65.65	48.41	**88.12**	56.67
	MFCC	40	62.75	71.15	51.45	84.93	63.04
	MFCC	80	75.79	**76.81**	40.94	58.09	45.37
	LFCC	20	60.72	53.04	54.78	59.28	63.33
	LFCC	40	63.47	75.50	62.32	77.25	69.28
	LFCC	80	66.95	71.73	54.06	83.04	56.09
	STFT	257	**84.05**	76.66	**66.38**	79.13	**73.77**

4.2 Transferability to Spoof Speech Detection

Table 4 presents the performance of different machine learning models for spoof speech detection on the ASVspoof2019 LA dataset using MFCC, LFCC, and STFT representations. Unlike the VAD experiments on MUSAN, where STFT achieved the strongest overall performance, LFCC emerged as the most effective representation for spoof speech detection. This shift suggests that the suitability of feature representations is task-dependent, further supporting the transferability analysis of the proposed framework.

RF and KNN demonstrated comparatively weaker performance, particularly in F1-score, suggesting reduced effectiveness under class imbalance. Overall, these findings indicate that the proposed CLBP-based framework is not limited to conventional VAD, but can generalize effectively to a related speech classification task with appropriate feature representations.

Among all evaluated configurations, XGBoost combined with LFCC-80 achieved the best overall performance, reaching 96.12% accuracy, 82.50% F1-score, and the lowest EER of 5.05%. This may be attributed to LFCC's ability to preserve linearly distributed spectral information, which can better expose subtle artifacts introduced by voice conversion and text-to-speech synthesis. The strong performance of XGBoost further suggests that non-linear ensemble learning is effective in exploiting discriminative patterns within the CLBP-enhanced feature space.

A clear performance trend can also be observed across feature representations. LFCC consistently outperformed MFCC across most classifiers, while STFT achieved competitive performance in several settings, particularly with SVM and LR. This contrasts with the VAD experiments, where STFT was generally the most robust representation, indicating that different speech analysis tasks may benefit from distinct spectral characteristics.

4.3 Comparison with previous studies

Figures 6 and 7 compare the proposed CLBP-based framework with representative VAD methods in terms of accuracy and inference time. In terms of classification accuracy, the XGBoost classifier using the proposed CLBP-based feature representation achieves approximately 99.8%, outperforming conventional handcrafted approaches reported by Dey et al. [3] (86.0%–89.0%). This improvement

Table 4: Performance of Machine Learning Models on ASVspoof 2019 LA (%)

Model	**Feature**	**numCoef**	**Accuracy**	**F1**	**EER**
XGBoost	MFCC	20	90.95 ± 0.2150	64.63 ± 0.6932	11.50 ± 0.2141
	MFCC	40	93.15 ± 0.1754	70.95 ± 0.6439	9.88 ± 0.3056
	MFCC	80	93.56 ± 0.1310	72.25 ± 0.5426	9.51 ± 0.2688
	LFCC	20	95.64 ± 0.1445	80.29 ± 0.4963	6.25 ± 0.0893
	LFCC	40	95.90 ± 0.0951	81.50 ± 0.3319	5.49 ± 0.0908
	LFCC	80	**96.12 ± 0.1249**	**82.50 ± 0.5112**	**5.05 ± 0.2335**
	STFT	257	94.45 ± 0.1250	75.86 ± 0.5308	7.53 ± 0.2237
RF	MFCC	20	92.56 ± 0.1230	48.59 ± 1.1227	13.76 ± 0.4218
	MFCC	40	93.49 ± 0.0999	55.58 ± 0.8934	12.75 ± 0.4183
	MFCC	80	92.84 ± 0.0844	49.49 ± 0.6859	13.25 ± 0.4509
	LFCC	20	94.66 ± 0.0876	66.34 ± 0.6866	9.43 ± 0.1398
	LFCC	40	93.82 ± 0.0435	58.21 ± 0.3178	9.03 ± 0.1420
	LFCC	80	93.85 ± 0.0346	59.31 ± 0.2885	8.04 ± 0.2246
	STFT	257	92.13 ± 0.0846	38.70 ± 0.9517	13.92 ± 0.3665
SVM	MFCC	20	91.70 ± 0.0742	47.93 ± 0.45	14.17 ± 0.4067
	MFCC	40	92.79 ± 0.0597	57.34 ± 0.5077	12.95 ± 0.2960
	MFCC	80	93.41 ± 0.1095	61.66 ± 0.6503	12.32 ± 0.2407
	LFCC	20	94.93 ± 0.1194	72.56 ± 0.6558	8.74 ± 0.1105
	LFCC	40	93.73 ± 0.0813	65.25 ± 0.2606	9.79 ± 0.2575
	LFCC	80	94.13 ± 0.0900	67.77 ± 0.5388	8.87 ± 0.1424
	STFT	257	95.21 ± 0.0454	74.18 ± 0.1850	8.42 ± 0.2304
KNN	MFCC	20	91.35 ± 0.0676	52.14 ± 0.7359	18.97 ± 0.3755
	MFCC	40	92.52 ± 0.1162	58.74 ± 0.6022	17.26 ± 0.2267
	MFCC	80	92.93 ± 0.0983	58.44 ± 0.6893	15.56 ± 0.2352
	LFCC	20	92.67 ± 0.1284	47.36 ± 1.0407	8.81 ± 0.2824
	LFCC	40	92.72 ± 0.0924	54.89 ± 0.5913	13.30 ± 0.1326
	LFCC	80	94.40 ± 0.0961	69.00 ± 0.4814	10.39 ± 0.1634
	STFT	257	91.83 ± 0.0681	47.10 ± 0.5159	17.28 ± 0.3829
LR	MFCC	20	85.01 ± 0.1787	53.66 ± 0.4078	14.21 ± 0.3785
	MFCC	40	86.25 ± 0.0868	56.09 ± 0.2962	13.05 ± 0.2731
	MFCC	80	87.20 ± 0.1128	57.86 ± 0.2882	12.40 ± 0.2447
	LFCC	20	90.78 ± 0.2623	66.53 ± 0.6292	8.81 ± 0.1381
	LFCC	40	89.31 ± 0.2107	63.11 ± 0.4571	9.84 ± 0.2608
	LFCC	80	90.27 ± 0.1890	65.45 ± 0.3501	8.94 ± 0.1355
	STFT	257	91.11 ± 0.2512	67.41 ± 0.6069	8.47 ± 0.2159

suggests that incorporating CLBP texture descriptors alongside acoustic features provides a more discriminative representation for VAD. Regarding inference efficiency, the *XGBoost + STFT + CLBP* configuration achieves 99.802% accuracy with an inference time of only 4.75 ms for a 2-second audio sample. Compared with recent lightweight deep learning methods, it is more than twice as fast as MagicNet [18] (12.07 ms) while achieving a 0.272% higher accuracy (99.802% vs. 99.530%). Compared with NASVAD [19] (19.91 ms, 99.830%), the proposed method reduces inference time by approximately 76% while maintaining a comparable accuracy with only a 0.028% difference. Although the *LR + MFCC20 + CLBP* configuration reduces inference time to 2.57 ms, its accuracy decreases to 98.750%, indicating a trade-off between computational cost and classification performance.

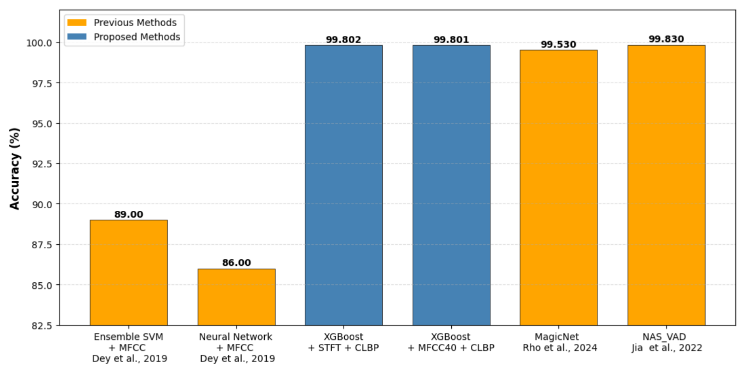

Fig. 6: Comparision of Accuracy on Musan Dataset

5 Conclusion

This study proposed a lightweight speech classification framework that integrates handcrafted spectral representations, CLBP encoding, and conventional ML classifiers. Experimental results on the MUSAN dataset demonstrated that the proposed framework achieves a favorable balance between detection performance and computational efficiency. In particular, the combination of XGBoost and STFT achieved 99.802% accuracy with an EER of 0.194%, while requiring only 4.75 ms to process a 2-second audio segment. Compared with recent DL methods, the proposed framework reduces inference latency by more than half while maintaining competitive detection accuracy. Cross-domain evaluations on the Aurora-2 dataset further demonstrated the robustness of STFT-based representations under unseen noisy conditions. In addition, experiments on the ASVspoof 2019 LA dataset showed that LFCC-based representations can be effectively transferred to spoof speech detection. These results suggest that combining local texture descriptors with conventional acoustic features provides an effective low-latency alternative to deep neural network-based approaches, making the proposed framework suitable for real-time and resource-constrained voice activity detection applications. Future work will investigate the applicability of the proposed framework to a broader range of speech and audio analysis tasks.

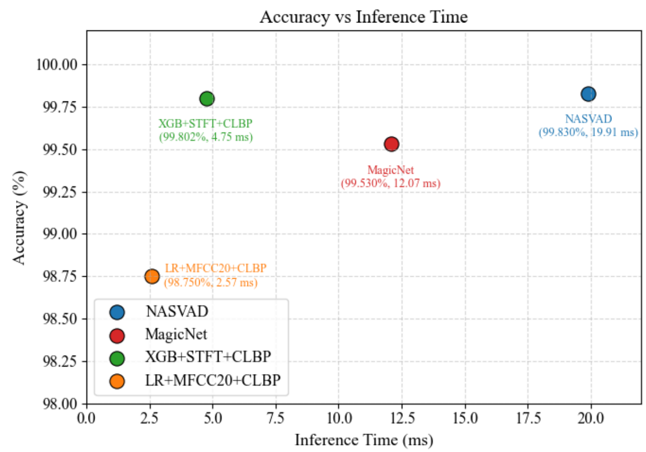

Fig. 7: Accuracy versus inference time for a 2-second audio sample. Inference time includes feature extraction and classification.

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